

Manuscript

Layer-1

稿件草稿 HTML 版（基于 docs/manuscript_draft.md）

Self-contained academic HTML with embedded SciencePlots figures (base64).

A literature-anchored Layer-1 surrogate for exploratory screening of indirect mitral annuloplasty

Short title: FMR IMA Layer-1 exploratory surrogate

Venue class: computational biomechanics / research-software methods (CMBBE / *Medical Engineering & Physics* / *Royal Society Open Science*-class). This is not a clinical decision trial.

One-sentence argument. We release a transparent, literature-anchored algebraic surrogate that reproduces selected Galili peak-systole anchors, incorporates Dryad-derived contact features, reports true fold-wise response-model cross-validation alongside a separate rule-based blend-off diagnostic, and ranks exploratory IMA-CS / IMA-AP scenarios under explicit geometric and transfer-efficiency assumptions—without claiming new Living Heart Human Model (LHHM) finite-element analysis (FEA), clinical recommendation, or patient-level external validation.

Reproducibility. `python run_pipeline.py --seed 42 --paper · python tools/import_galili_dryad.py --from-fixture --process · python -m analysis.fit_response_model --write · python tools/loo_evaluate.py --write`

Verification levels. Level 0 = calibrated to / reproduced selected Galili cases. Level 1 = exploratory planning map + uncertainty-aware scenario ranker + fold-wise response diagnostic. Level 2 patient-specific LHHM / fluid–structure interaction (FSI) is out of scope.

Abstract

Background. Computational studies of indirect mitral annuloplasty (IMA) report suture or bridge shortening together with peak-systolic geometry and leakage. Galili et al. (*R. Soc. Open Sci.* 2022; doi:10.1098/rsos.211464) compared generic coronary-sinus (IMA-CS) and anteroposterior (IMA-AP) strategies in the LHHM. Peak-systolic IMA-AP at 50% shortening yields an anteroposterior (AP) diameter of 15.9 mm (near-direct AP effect relative to disease 26.1 mm), which must not be confused with undeformed diastolic AP of 34.4 mm. Device classes also differ clinically: MAVERIC evaluates ARTO (CS–interatrial septum / IMA-AP-class), whereas Carillon and REDUCE-FMR are IMA-CS-class.

Methods. We implement a reproducible Python Layer-1 surrogate that couples phenomenological mechanics with a literature-calibrated leakage proxy. A configuration switch selects the response path used for ranking (`fitted_response` is the seed-42 paper default; `rule_based_proxy` is retained for diagnostics). Dryad supporting files (doi:10.5061/dryad.bzkh1899d) supply provenance and an auxiliary feature `contact_fraction` (interpolated off-table); scalar AP, regurgitant orifice area (ROA), and leakage remain table-backed unless independently derivable. Fold-wise response models $(f_{\{\mathrm{ROA}\}}(\Delta\mathrm{AP}, \mathrm{device\ family}, \mathrm{annular\ reduction}, \mathrm{contact_fraction}))$ and $(f_{\{\mathrm{leak}\}}(\mathrm{ROA}, \mathrm{coaptation\ proxy}, \mathrm{contact}, \mathrm{device\ mechanism}))$ are fitted under true leave-one-case-out; a full-train fit supports ranking. A discrete scenario ranker minimizes `leakage_proxy_pct` subject to an AP-reduction ceiling, an illustrative nickel–titanium (NiTi) engineering screen, and a Rottländer coronary sinus–left circumflex (CS–LCx) risk-screening threshold. Latin-hypercube sampling (LHS; $N=120$) probes multi-parameter ranking stability. The dual-suture commissural factor is treated as hypothesis sensitivity (Figure 5), not discovery.

Results (seed = 42). On the held-out Galili subset, fold-wise response-model cross-validation (CV) achieves ROA mean absolute error (MAE) $\approx 6.37 \text{ mm}^2$ and leak MAE ≈ 0.245 percentage points (pp), versus a documented pre-dampen rule-based blend-off baseline of $50.18 \text{ mm}^2 / 1.445 \text{ pp}$. For IMA-AP 70%, fold-wise absolute errors fall to $\approx 6.60 \text{ mm}^2 / 0.198 \text{ pp}$ (internal engineering screens only—not medical validation). The current dampened rule-based diagnostic yields held-out MAE $\approx 5.24 \text{ mm}^2 / 0.043 \text{ pp}$; for IMA-AP 70% it predicts leak $\approx 0.07\%$ against published 0.13% (absolute error $\approx 0.06 \text{ pp}$). Paper ranking therefore uses `fitted_response` for a consistent off-table response surface, not because every scalar is worse under the rule-based path. Under the planning map the ranker evaluates 36 grid points (35 device candidates) and retains 30 feasible designs ($p_{\{\mathrm{feasible}\}} \approx 0.857$). The best feasible candidate is IMA-CS bridge shortening 20% (AP reduction 11%; leakage proxy $\approx 0.39\%$; jet = central; CS–LCx = 8.6 mm at the risk-screen edge). LHS uncertainty quantification (UQ) yields $(P(\mathrm{top\text{-}1}) \approx 0.34)$ and mean feasible fraction ≈ 0.82 .

Conclusions. A literature-anchored low-order surrogate can support exploratory screening of IMA strategy settings with inspectable assumptions, honest held-out reporting, and uncertainty language. It is not a preoperative clinical decision system and must not equate physics leakage-proxy percentages with clinical regurgitant volume.

Keywords: functional mitral regurgitation; indirect mitral annuloplasty; surrogate model; exploratory screening; research software; Layer-1

中文摘要. 背景：功能性二尖瓣反流（FMR）的间接成形（IMA）计算研究需区分峰缩期前后径（AP）与舒张期几何，并区分 IMA-CS 与 IMA-AP 装置类别。方法：本文给出可复现的 Python 一层代数/现象学代理，结合 Dryad 接触特征、折内响应模型交叉验证与情景排序。结果（seed=42）：折内留出子集 ROA MAE≈6.37 mm²、泄漏 MAE≈0.245 pp；排序最优候选为 IMA-CS 20%（AP↓11%，泄漏代理≈0.39%）。结论：可用于假设透明的探索性筛查，而非临床术前决策或患者级外部验证。

1. Introduction

Functional mitral regurgitation (FMR) arises from ventricular remodelling, annular dilatation, and impaired leaflet coaptation rather than primary leaflet disease alone. Indirect mitral annuloplasty (IMA) seeks to remodel annular geometry without direct leaflet repair. Two computational device families recur in the literature: IMA-CS (coronary-sinus bridge shortening; Carillon-class clinically) and IMA-AP (CS–interatrial-septum suture shortening that acts primarily on the anteroposterior diameter; ARTO / MAVERIC mechanism class). These pathways are not mechanically interchangeable.

Galili, White Zeira and Marom (2022) compared generic IMA-CS and IMA-AP treatments in the Living Heart Human Model and quantified leakage after device deployment. Their peak-systolic IMA-AP 50% case yields AP = 15.9 mm, illustrating a near-direct AP effect; undeformed diastolic AP = 34.4 mm belongs to a different cardiac phase and must not be substituted into peak-systole dose statements. Clinical series supply geometric context at different scales: ARTO/MAVERIC reports AP reductions of order 14–15% for an IMA-AP-class device, whereas Carillon / TITAN II and REDUCE-FMR inform IMA-CS directionality.

What remains for research software is transparent dose translation, provenance, held-out honesty, and exploratory ranking under stated assumptions. The present work does not claim the first computational CS-versus-AP comparison, does not introduce new LHHM FEA, and does not offer a clinical recommendation engine.

2. Related work and positioning

We imitate the *structure* of high-fidelity IMA modelling papers and modular mitral-valve methods tools while contributing a complementary Layer-1 software layer:

Role	Source	What we imitate	What we do not claim
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Mechanistic FEA baseline	Galili et al., <i>R. Soc. Open Sci.</i> 2022;9:211464	mechanism → parameterization → geometry → functional readout; IMA-CS vs IMA-AP taxonomy	new LHHM/FEA/SPH; “first CS vs AP”
Clinical AP context	Worthley / MAVERIC (ARTO)	separate geometry endpoints from MR endpoints	calibration of physics % to clinical MR volume
CS directionality	Witte et al., REDUCE-FMR; Carillon TITAN II	restrained causal language; coronary awareness	validation against trial magnitudes
LCx screening vocabulary	Rottländer et al., 2021	feasible vs infeasible screening language at CS–LCx $\approx$ 8.6 mm	“ $\geq$ 8.6 mm is proven safe”
Methods-tool tone	geometry-based MV FE tools (e.g. <i>Med Eng Phys</i> / <i>CMBBE</i> modular studies)	modular Methods; isolated-variable sweeps	FE stress/contact as ground truth for our proxies

This paper contributes research-software packaging, Dryad-aware contact features in a fitted response path, fold-wise response diagnostics, and an inspectable Layer-1 scenario ranker—not a new high-fidelity FSI study.

3. Software architecture

```

configs/      surrogate_calibration.yaml, design_space.yaml, case YAML
models/      geometry, pathology, devices, response_model (fold-wise f_ROA / f_leak)
simulation/  phenomenological mechanics proxy (legacy alias run_fea_surrogate)
sph/        literature-calibrated leakage proxy (SPH-inspired; not a full SPH solver)
analysis/    ROA, jet, sweep, scenario_ranker, fit_response_model, paper tables/plots
tools/       import_galili_dryad.py, loo_evaluate.py, package_reports.py
data/       raw drop zone, fixtures, processed galili_cases.csv
results/output/cross_validation/  true fold-wise CV artifacts
results/output/planner/          scenario_ranking.json (seed-42 ranking)

```

Entry points include `run_pipeline.py`, `python -m analysis.planner`, `python -m analysis.fit_response_model`, and `python -m analysis.design_sweep`.

4. Methods

4.1 Pathology and geometry

A parametric papillary pathology (approximately 44% posterior passive fraction) defines the diseased baseline. Undeformed diastole and peak systole are stored as separate `cardiac_phase` records. AP diameter is a literature-defined geometry input (`ap_role: literature_mapping_input`); AP MAE is therefore not reported as predictive accuracy.

4.2 Device kinematics and transfer efficiency

In Galili mapping mode, peak-systolic AP follows published table coordinates (interpolation between discrete Galili cases). In clinical / planning mode, AP reduction is $\mathrm{AP_reduction\%} = \eta \times \mathrm{shortening\%}$ with assumption-prior means $\eta_{\mathrm{AP}} = 0.30$ and $\eta_{\mathrm{CS}} = 0.55$. η_{CS} is **not** fitted from MAVERIC/ARTO. Default UQ uses Latin hypercube sampling over $\eta_{\mathrm{AP}} \in U(0.24, 0.36)$, $\eta_{\mathrm{CS}} \in U(0.44, 0.66)$, dual commissural factor, CS–LCx baseline, and cinch slope ($N=120$) as configured in `design_space.yaml`.

4.3 Mechanics and leakage proxies

Algebraic coaptation-gap, strain, and contact-proxy scores constitute the phenomenological mechanics channel. The leakage proxy is literature-calibrated at the pathology case; physics regurgitation percentage is **not** clinical regurgitant volume. Synthetic contact-cluster ROA is visualization-only (`model_weight=1`, `cluster_weight=0`). Dryad `contact_fraction` enters the fitted response path as a scientific feature. Official export fields are `leakage_proxy_pct`, `strain_risk_score`, and `contact_score` (legacy aliases retained for compatibility).

4.4 Response paths

`configs/design_space.yaml` sets `response_path` to one of:

- `fitted_response` — paper / seed-42 default for ranking;
- `rule_based_proxy` — legacy exploratory / diagnostic path;
- `hybrid_ap_extreme` — rule-based with AP-class over-short routed to the fitted surface.

4.5 Provenance (Dryad)

`tools/import_galili_dryad.py` documents download or manual drop, then extracts `contact_fraction` and `n_sph_particles` where available. Scalar AP, ROA, and leakage remain table-backed fills; the importer does not invent those scalars from unlabeled leaflet coordinates alone.

4.6 Diagnostics versus true fold-wise CV

tools/loo_evaluate.py runs an **anchor-free / leave-one-case blend-off diagnostic** on the fixed rule-based surrogate (no fold-wise refit). analysis/fit_response_model.py performs **true fold-wise** fitting with held ROA/leak excluded from training arrays. Full-train parameters for ranking are written under results/output/response_model/. Cross-validation summaries live under results/output/cross_validation/.

4.7 Scenario ranker and Pareto families

The ranker minimizes leakage_proxy_pct subject to an exploratory AP-reduction ceiling (default 20%), an illustrative NiTi alternating-strain screen ($<0.4\%$), and a CS–LCx risk-screening threshold (≥ 8.6 mm; labeled screening, not safety clearance). It reports n_total_points, n_device_candidates, n_feasible_device_candidates, p_feasible_device_candidates, $\backslash (P(\mathrm{top}\text{-}1))$, and first-order Spearman sensitivity ranks. Pareto families are separated: global common objectives (min leak, max AP $\downarrow$); IMA-CS family (adds LCx/NiTi); IMA-AP family. Missing LCx/NiTi values are never replaced with sentinel fillers. The dual-suture commissural factor is an explicit hypothesis sensitivity parameter (Figure 5: 0.25 / 0.5 / 0.75 / 1.0). The 14–20% AP band is a clinically contextualized exploratory planning range (legacy clinical_window alias only).

5. Results

5.1 Data provenance

Processed table data/processed/galili_cases.csv marks Dryad-derived contact_fraction and n_sph_particles with source_tier=dryad_derived. Published peak-systole AP, ROA, and leakage scalars remain table-backed.

5.2 Calibration reproduction (Level 0)

Case	Leakage %	Peak-sys AP mm	ROA mm ²
pathology	5.26	26.1	172.8
ima_cs_22	0.29	24.8	48.2
ima_ap_50	0.08	15.9	27.3

ima_ap_70	0.13	12.4	46.1
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Calibration IDs with blend ON are reproduction only.

5.3 Anchor-free diagnostic and true fold-wise CV

Rule-based blend-off diagnostic (`tools/loo_evaluate.py`). A documented pre-dampen historical baseline on the held-out IDs reported MAE ROA 50.18 mm² and leak 1.445 pp, with IMA-AP 70% absolute errors 94.17 mm² / 5.455 pp. After rule-based dampening, the current seed-42 diagnostic reports held-out MAE $\approx$ 5.24 mm² / 0.043 pp. For IMA-AP 70%, current rule-based leak $\approx$ 0.07% versus published 0.13% (absolute error $\approx$ 0.06 pp).

True fold-wise response-model LOO (`analysis/fit_response_model.py`). Held-out-subset MAE ROA $\approx$ 6.37 mm² and leak $\approx$ 0.245 pp; IMA-AP 70% absolute errors $\approx$ 6.60 mm² / 0.198 pp (pred ROA 39.5 mm² vs 46.1; pred leak 0.33% vs 0.13%). Internal engineering screens (ROA MAE < 25, leak MAE < 0.5, AP70 ROA < 25, AP70 leak < 0.5) are met on this seven-case table—not patient-level external validation. Full seven-fold MAE remains inflated by the pathology leave-out (large ROA scale gap). AP prediction metrics remain inapplicable.

Held-out table (CS14 / CS18 / AP30 / AP70) — published versus current rule-based blend-off versus fold-wise response model (seed-42 pipeline outputs):

Case	Pub ROA	Rule ROA (abs err)	Fold ROA (abs err)	Pub leak %	Rule leak (abs err pp)	Fold leak (abs err pp)
ima_cs_14	56.7	59.6 (2.9)	69.6 (12.9)	0.52	0.51 (0.01)	0.60 (0.08)
ima_cs_18	55.3	54.9 (0.4)	53.4 (1.9)	0.41	0.42 (0.01)	0.42 (0.01)
ima_ap_30	51.1	42.4 (8.7)	47.0 (4.1)	0.16	0.25 (0.09)	0.86 (0.70)
ima_ap_70	46.1	37.2 (8.9)	39.5 (6.6)	0.13	0.07 (0.06)	0.33 (0.20)

Paper ranking prefers `fitted_response` for a coherent off-table surface under planning-map kinematics; the rule-based path remains the named blend-off diagnostic.

5.4 Exploratory scenario ranking (planning map, seed = 42; fitted_response)

- $(n_{\text{total}}=36)$, $(n_{\text{device candidates}}=35)$, $(n_{\text{feasible}}=30)$, $(p_{\text{feasible}})\approx 0.857$

- Best feasible candidate: **IMA-CS bridge 20%** ($\eta_{\mathrm{CS}}=0.55$) $\rightarrow$ AP reduction **11.0%**, leakage proxy $\approx$ **0.391%**, jet = central, CS–LCx **8.6 mm**)
- Best IMA-AP single / dual alternatives under the same constraints remain higher-leakage ($\approx$ 0.79% at 65% shortening / 19.5% AP $\downarrow$); dual suture is reported as hypothesis sensitivity (Figure 5), not as Innovation D discovery
- LHS multi-parameter UQ ($N=120$): $P(\mathrm{top}\text{-}1) \approx 0.342$, mean feasible fraction ≈ 0.822 ; first-order Spearman ranks are dominated by CS–LCx baseline and cinch slope
- $\pm 20\%$ one-at-a-time sensitivity keeps IMA-CS 20% as the nominal top candidate while shifting realized AP reduction (8.8% / 11.0% / 13.2%)

5.5 Directionality-only clinical context

ARTO/MAVERIC: AP $\downarrow$ of order 14–15% (IMA-AP class). Carillon TITAN II: $\sim$ 15% AP context (IMA-CS class). REDUCE-FMR: regurgitation $\downarrow$ directionality. Magnitudes are not equated to physics leakage-proxy percentages.

6. Discussion

The central contribution is transparent exploratory screening software: corrected Galili peak-systole AP semantics, MAVERIC = ARTO device-class attribution, Dryad `contact_fraction` in the fitted path, honest held-out / CV reporting (including IMA-AP 70%), LHS ranking stability, and family-correct Pareto language.

Relative to Galili’s LHHM study, we do not re-litigate which generic device seals better in high fidelity. Relative to clinical series, we borrow geometric windows and screening vocabulary only. Relative to modular mitral FE tools, we trade continuum fidelity for inspectable assumption priors and rapid grid ranking.

Limitations (documented, not papered over):

- $n=7$ Galili peak-systole table cases — not patient-level external validation
- No patient CT / patient-specific chamber geometry
- No chamber-labeled SPH leakage recompute from Dryad particle clouds (full-domain cloud only; AP/ROA/leak remain table-backed)
- Algebraic / phenomenological proxies; η and dual factor are assumption priors
- LCx/NiTi screens are literature / engineering screens, not safety clearances
- Ranking is assumption-driven exploratory analysis, not validated prediction

7. Availability

- Code: <https://github.com/Coucou2016/fmr-ima-layer1-planner>
- License: MIT (LICENSE)
- Citation: CITATION.cff · references.bib
- Model card: MODEL_CARD.md · Data: DATA_PROVENANCE.md

8. Conclusion

A Layer-1 literature-anchored surrogate can rank exploratory IMA scenarios under explicit assumptions after correcting peak-systole Galili AP facts and MAVERIC device-class attribution, and after reporting true fold-wise response diagnostics alongside a renamed blend-off casewise diagnostic. Cite as methods / research software for exploratory screening—not as clinical preoperative decision support.

Figure captions (SciencePlots; English axis labels)

Figure 1. IMA-AP leakage proxy versus suture shortening under Galili peak-systole AP mapping and clinical η mapping. The shaded band marks the exploratory 14–20% AP planning range expressed in suture-% space via η_{AP} .

Figure 2. Device shortening mapped onto AP-diameter reduction for Galili IMA-AP, clinical IMA-AP ($\eta=0.30$), and clinical IMA-CS ($\eta=0.55$) assumption prior). Horizontal band: exploratory planning range.

Figure 3. Commissural fraction of ROA and jet-location labels versus shortening (Galili vs clinical panels; squares = IMA-AP, circles = IMA-CS).

Figure 4. IMA-CS clinical-map Pareto / risk-screen view: leakage proxy versus CS–LCx distance and versus NiTi alternating strain. Vertical dashed lines mark the Rottländer 8.6 mm screen and the 0.4% engineering strain screen.

Figure 5. Dual-suture commissural-factor sensitivity (0.25 / 0.5 / 0.75 / 1.0) versus single-suture baseline—hypothesis parameter, not discovery.

Embedded SciencePlots figures

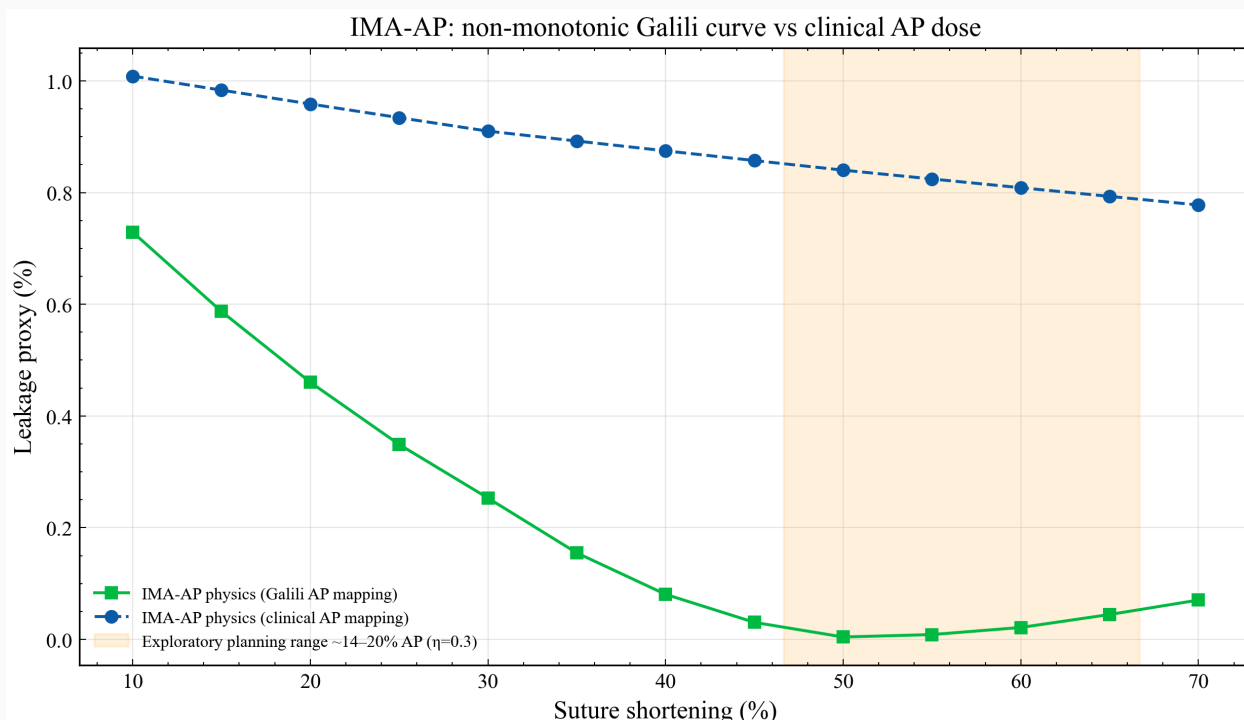


图 1. IMA-AP 泄漏代理随缝线缩短变化：Galili 峰缩期映射 vs 临床 η 映射

文件: fig1_ima_ap_nonmonotonic_exploratory_planning_range.png (SciencePlots + Times New Roman, dpi $\geq$ 300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉 (为什么画这张图): 间接二尖瓣成形 (IMA, indirect mitral annuloplasty) 文献常同时报告“缝线缩短百分比”

与泄漏。读者容易把计算缝线 % 直接当成临床前后径 (AP, anteroposterior diameter) 缩减 %。本图把同一套

Layer-1 代理放在两条映射曲线上, 用来回答: 缩短 % 与泄漏代理之间的关系如何随几何映射改变?

坐标与量的定义: 横轴 = IMA-AP 缝线缩短 %; 纵轴 = leakage_proxy_pct (泄漏代理百分比)。该量来自代数/现象学

力学通道与文献校准泄漏代理 (或 fitted_response 路径), 不是超声测得的反流容积分数, 也不是临床试验主要终点。

如何读: (1) Galili 映射复现发表峰缩期几何: IMA-AP 50% $\rightarrow$ 峰缩期 AP=15.9 mm (疾病峰缩期 26.1 mm; 近直接缩 AP), 切勿写成未变形舒张期 34.4 mm / “0% AP”。70% 进一步塌缩 AP 并在代理上显示交界区泄漏风险升高的非单调倾向。

(2) 临床/规划映射采用假设先验 AP_reduction% = $\eta \times$ shortening%, 默认 $\eta_{AP}=0.30$ (假设先验, 非成像-力学辨识,

亦非 MAVERIC 拟合)。绿色带为 exploratory planning range, 把约 14–20% AP 对话框口反投

到缝线 % 轴。

与 seed-42 排序的关系：本图解释剂量语义，不单独给出“最优装置”。seed-42 在 fitted_response + 临床映射下

最优候选是 IMA-CS 20%（见图 4 / 表 1），IMA-AP 曲线在此图中主要用于对照映射约定。

诚实边界：纵轴 $\neq$ 临床反流容积；Galili 峰缩期事实来自 doi:10.1098/rsos.211464；本仓库不新跑 LHHM/Abaqus。

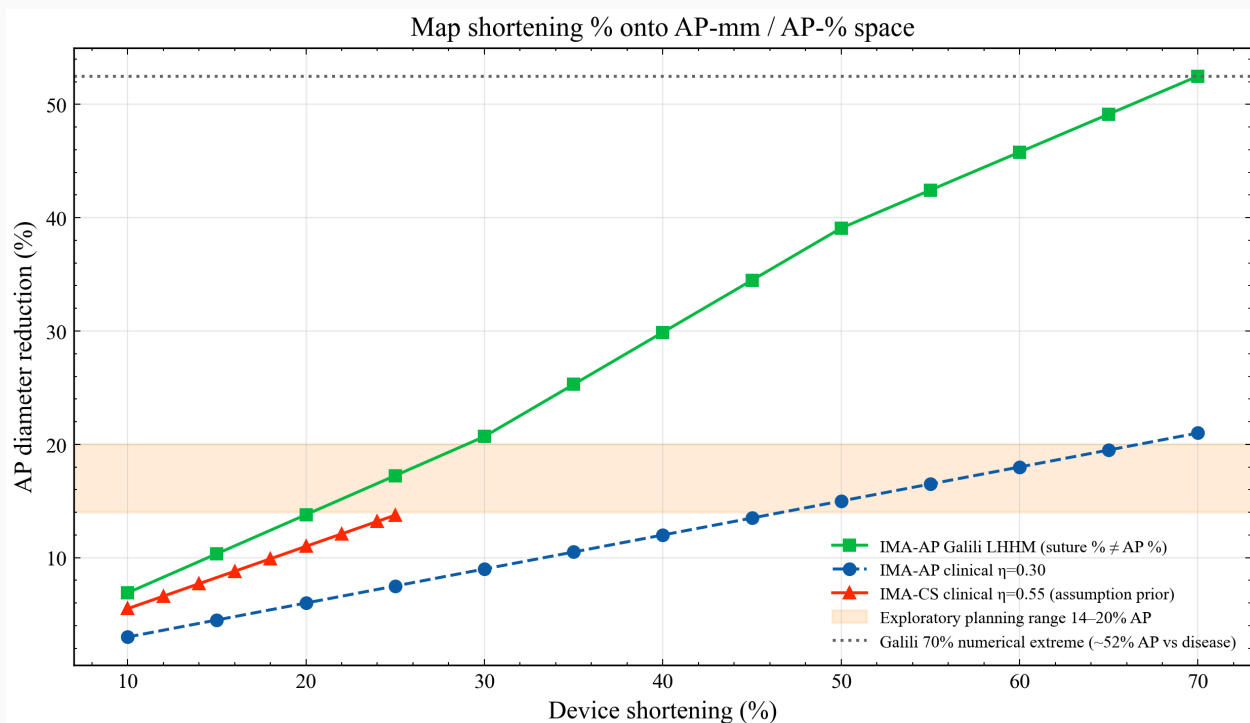


图 2. 装置缩短百分比到 AP 直径缩减百分比的剂量映射

文件: fig2_suture_vs_ap_reduction.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉: 图 1 看“缩短→泄漏”, 本图看“缩短→AP 缩减”, 把剂量翻译问题单独拆开, 避免把泄漏变化误读成 AP 剂量。

坐标: 横轴 = 装置缩短 % (IMA-AP 缝线或 IMA-CS 桥); 纵轴 = AP 直径缩减 %。三条轨迹: (i) Galili 峰缩期 IMA-AP

(近直接缩 AP); (ii) 规划 IMA-AP ($\eta_{AP}=0.30$); (iii) 规划 IMA-CS ($\eta_{CS}=0.55$, 假设先验, 不是从 MAVERIC/ARTO

回归得到)。水平带 14–20% 为探索性规划窗口; MAVERIC=ARTO 属 IMA-AP 类语境, Carillon/TITAN II 属 IMA-CS 语境。

如何读: 同一缩短 % 在不同映射下对应的 AP% 可以相差很大。这正是 Layer-1“翻译层”存在的理由: 计算缝线坐标

与临床 AP 对话窗口不是同一坐标。点线“数值极端”标出 Galili 高剂量 IMA-AP 相对疾病的大幅 AP 塌缩, 提醒勿把

70% 缝线写成“临床缩 AP 七成”。

与结果衔接: seed-42 最优 IMA-CS 20% 在 $\eta_{CS}=0.55$ 下对应 $AP \downarrow \approx 11\%$ (低于 14–20% 窗口下沿, 但仍在 AP 上限 20%

与 LCx/NiTi 约束内因泄漏代理更低而胜出)。窗口是语境带, 不是硬可行域的唯一定义。

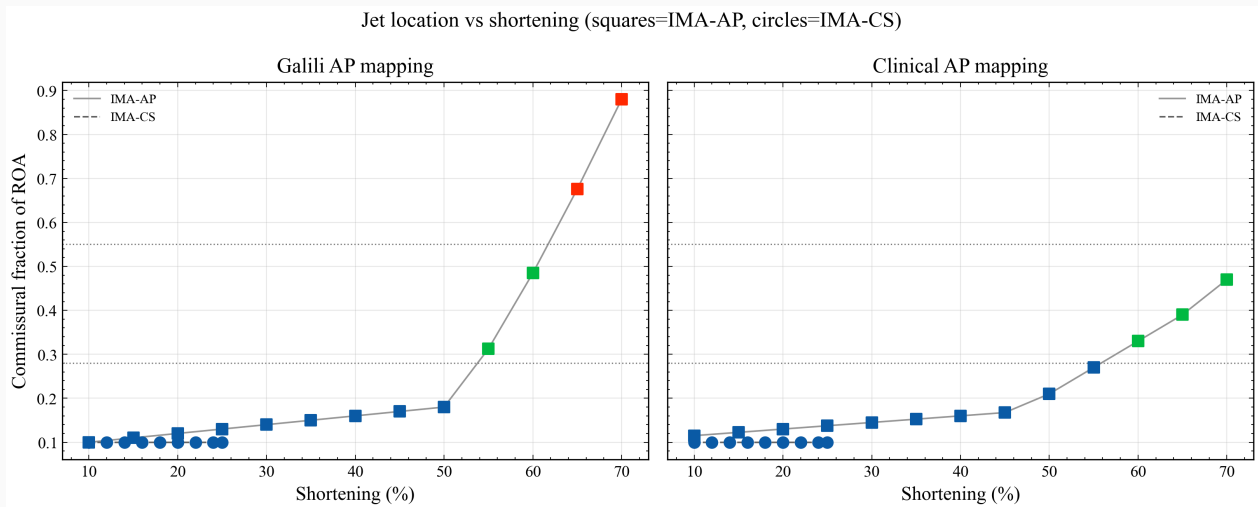


图 3. 射流位置与交界区 ROA 分数随缩短变化（Galili vs 临床）

文件: fig3_jet_location.png（SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明）

来龙去脉：除泄漏“量”外，筛查还关心泄漏机制位置。ROA（regurgitant orifice area，反流口面积， mm^2 ）在本代理中

由对合间隙等代数数量估计，并拆成中央份额与交界区份额；再按阈值分为 central / mixed / commissural。本图把该

机制标签随缩短 % 的变化画出来，左右对照 Galili 与临床映射。

读图约定：横轴 = 缩短 %；纵轴 = 交界区 ROA 分数；方块 = IMA-AP，圆点 = IMA-CS；水平虚线为分类阈值示意。

标签是代理口上的机制草图，不是超声 PISA（proximal isovelocity surface area）或多普勒诊断结论。

如何联系 seed-42：在 fitted_response 排序下，最优候选 IMA-CS 20% 的 jet='central'（交界分数约 0.10）。IMA-AP

高剂量单缝线在临床映射上更易进入 mixed；双缝线交界因子则在图 5 单独做假设敏感性，不在本图宣称“发现更优术式”。

结论（代理内）：机制标签随映射与剂量变化；可与表 1 / 图 5 交叉阅读，但不得把 jet 分类写成临床影像学验证。

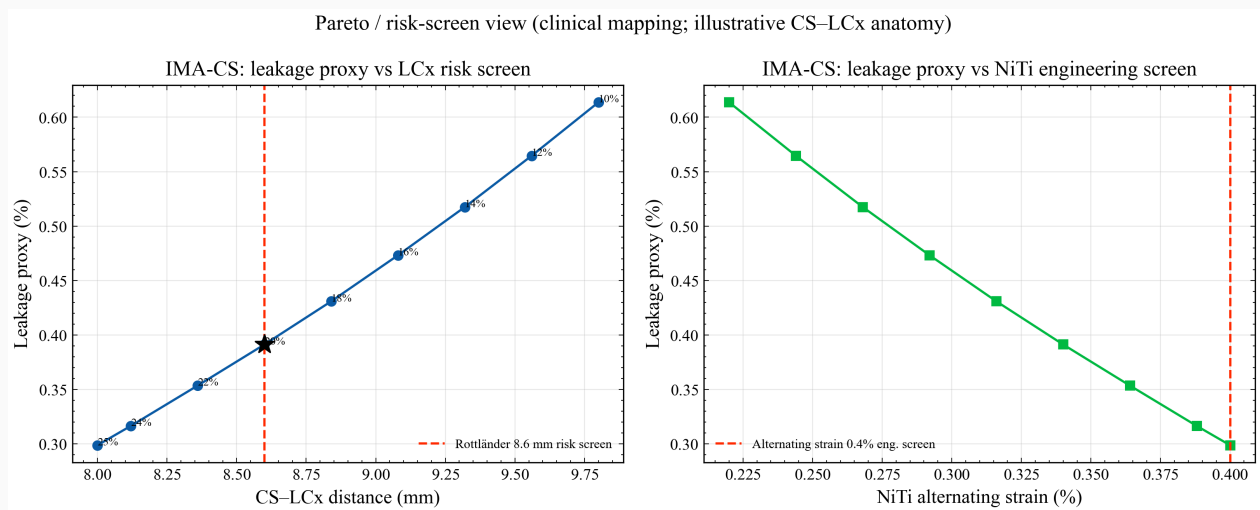


图 4. IMA-CS 临床映射：泄漏代理-筛查 Pareto（CS-LCx 与 NiTi 交变应变）

文件：fig4_pareto_lcx_strain.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉：IMA-CS（经冠状窦路径）在减小泄漏代理的同时，可能缩短 CS-LCx（coronary sinus-left circumflex artery，

冠状窦与左回旋支）距离。文献风险筛查常用远端着陆区 CS-LCx 约 8.6 mm 作为“预测冠状动脉妥协”的 ROC 语境阈值

（Rottländer 等）。同时对 NiTi（镍钛）桥交变应变设置 <0.4% 的工程筛查上限。默认示意解剖基线 CS-LCx=11.0 mm

（design_space 假设，可换成患者 CT）。标题语言是“筛查”，不是“安全证明”。

读图：左面板横轴 CS-LCx (mm)、纵轴泄漏代理 %；右面板横轴 NiTi 交变应变 %、纵轴泄漏代理 %。点随桥缩短移动；

越贴近约束边界，可行域越窄。星号（若出现）标记 seed-42 情景排序在假设下的最优 IMA-CS 候选。

与 seed-42 的直接对应：最优候选为桥缩短 20%，AP↓11%，泄漏代理≈0.39%，CS-LCx=8.6 mm（贴边），NiTi 交变应变

≈0.34%。更高桥缩短常因 CS-LCx <8.6 mm 被判不可行。这是网格筛查信号，不是患者级安全性结论。

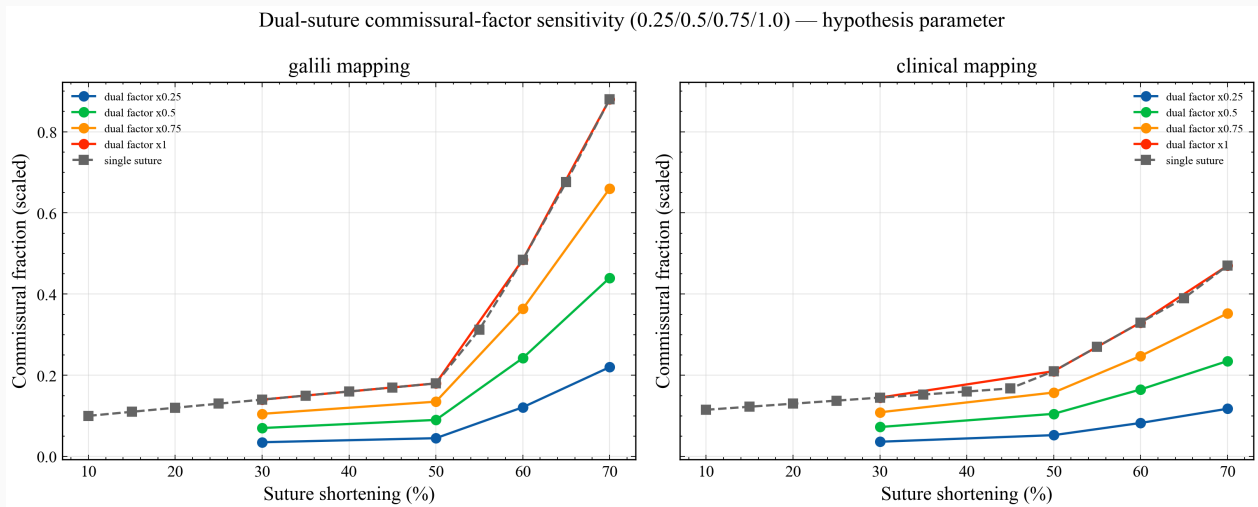


图 5. 双缝线交界因子敏感性（0.25/0.5/0.75/1.0）——假设参数，非发现声称

文件：fig5_dual_vs_single_suture.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉：双缝线（dual suture）在匹配 AP 缩减下可能改变交界区 ROA 份额。本仓库把“交界因子”写成显式假设参数，并在 0.25/0.5/0.75/1.0 上做敏感性，而不是把它包装成已证实的术式创新（禁止 Innovation D / discovery 话术）。

读图：左右为 Galili / 临床映射；曲线为不同因子下的交界分数；虚线为单缝线对照。名义因子 ×0.5 仅是默认假设。

与主排序的关系：seed-42 默认 fitted_response 排序的第一名是 IMA-CS 20%，不是双缝线 60%。本图回答“若坚持双缝假设，机制标签如何随因子变”，用于诚实展示假设敏感性；不得把图中 physics/机制差写成临床反流容积改善。